GETEK Materials

Low Dk/Df Laminate and Prepreg

GETEK materials provide the low dielectric constant (Dk) and low dissipation factor (Df) performance demanded by high speed, low loss printed wiring board (PWB) designs and applications while providing superior thermal performance and high reliability based on the systems 180°C glass transition temperature (Tg). GETEK laminate and prepreg products are manufactured using a functionalized, Polyphenylene Oxide/Epoxy resin, reinforced with electrical grade (E-glass) glass fabric. In addition to this superior electrical and thermal performance the mechanical, chemical and low moisture absorption properties all equal or exceed the performance of traditional FR-4 materials. The GETEK system is also UV blocking and fluorescing.

- **Industry Approvals**
  IPC–4101C /25
  UL Recognized – FR-4, File Number E41625
  (ML200+ and RG200+ listing)
  Qualified to UL's MCIL Program

- **High Thermal Performance**
  Tg of 180C (DMA)
  Low CTE for reliability

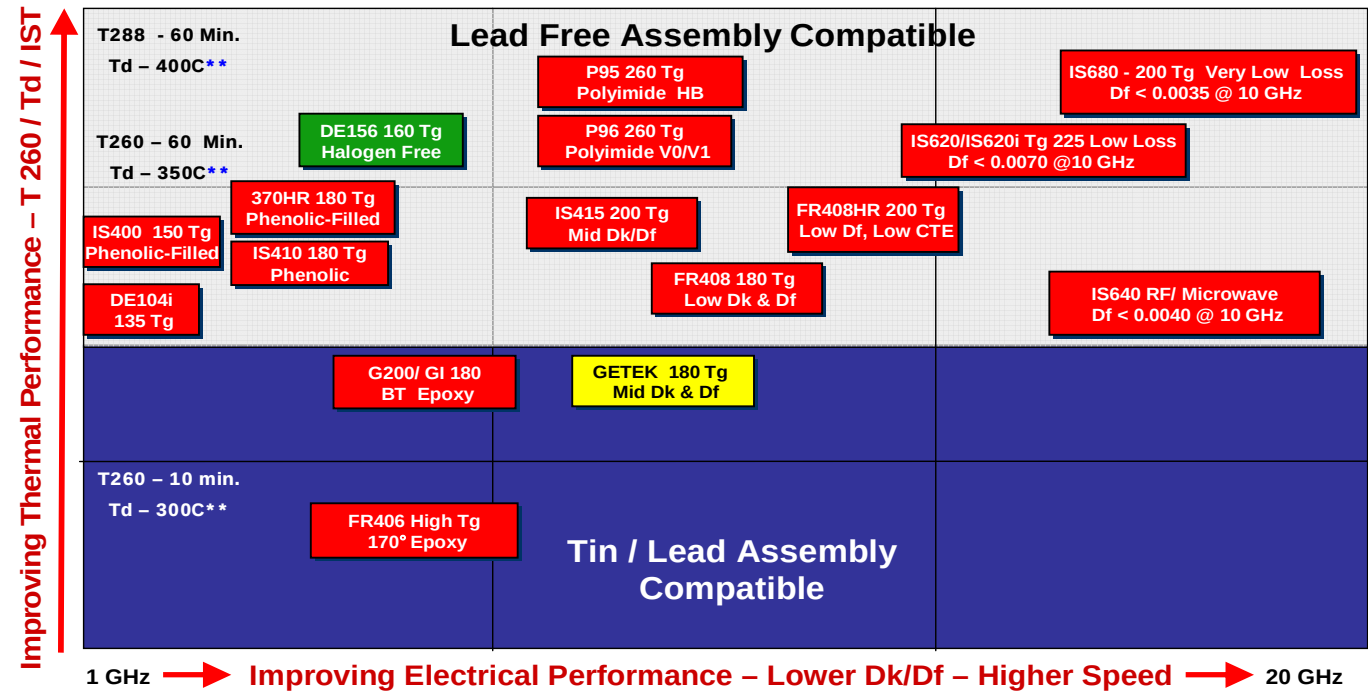
- **Improved Dielectric Properties**
  DK <3.8 (50MHz – 1GHz) - Supports increased signal speeds
  DF <0.010 (50MHz – 1GHz) – Provides better signal integrity

- **UV Blocking and AOI Fluorescence**
  High throughput and accuracy during PCB fabrication and assembly

- **Superior Processing**
  Closest to conventional FR-4 processing of all high speed materials

- **Standard Availability**
  Thickness: 0.002 [0.05 mm] to 0.093" [2.4 mm]
  Available in sheet or panel form
  Copper Foil Cladding: Grade 3 (HTE), ½ , 1 and 2 oz.
  Foil Options: Reverse treat
  Prepregs: Available in roll or panel form
  Glass Styles: Standard fabrics

Isola - Product Position
Thermal Performance vs Signal Integrity



Speed is a function of design such as line length etc.

** Laminate Data - IST performance is a function of Hole diameter, board thickness, plating parameters and laminate attributes.

GETEK Typical Laminate Properties

GETEK					
Property		Typical Values			
		Typical Value	Specification	Units Metric (English)	Test Method IPC-TM-650 (or as noted)
Glass Transition Temperature (Tg) by DSC, spec minimum		180	170-200	°C	2.4.25
Decomposition Temperature (Td) @ 5% wt loss		345	—	°C	ASTM D3850
CTE, Z-axis	A. Pre-Tg	55	AABUS	ppm/°C	2.4.24
	B. Post-Tg	275	—		
CTE, X-, Y-axes	A. Pre-Tg	13/14	AABUS	ppm/°C	2.4.24
	B. Post-Tg	14/17	—		
% Z-Axis Expansion (50-260C)		3.01		%	2.4.24
Thermal Conductivity		0.4	—	W/mK	ASTM D5930
Thermal Stress 10 Sec	A. Unetched	pass	Pass Visual	Rating	2.4.13.1
@ 288°C (550.4°F), spec min	B. Etched	pass	Pass Visual		
Permittivity, spec maximum (Laminate & prepreg as laminated)	A. @ 100 MHz HP4285A	3.81	5.4	—	2.5.5.3
	B. @ 1 GHz HP4291A	3.78	—		2.5.5.9
	C. @ 2 GHz Bereskin Stripline	3.60	—		2.5.5.5
	D. @ 5 GHz Bereskin Stripline	3.50	—		2.5.5.5
	E. @ 10 GHz Bereskin Stripline	3.50	—		2.5.5.5
Loss Tangent, spec maximum (Laminate & prepreg as laminated)	A. @ 100 MHz HP4285A	0.0110	0.035	—	2.5.5.3
	B. @ 1 GHz HP4291A	0.0100	—		2.5.5.9
	C. @ 2 GHz Bereskin Stripline	0.0090	—		2.5.5.5
	D. @ 5 GHz Bereskin Stripline	0.0090	—		2.5.5.5
	E. @ 10 GHz Bereskin Stripline	0.0100	—		2.5.5.5
Volume Resistivity, spec minimum	A. 96/35/90		1.0×10^6	MΩ -cm	2.5.17.1
	B. After moisture resistance	3.81×10^8	—		
	C. At elevated temperature	3.9×10^8	1.0×10^3		
Surface Resistivity, spec minimum	A. 96/35/90		1.0×10^4	MΩ	2.5.17.1
	B. After moisture resistance	2.81×10^6	—		
	C. At elevated temperature	2.64×10^8	1.0×10^3		
Dielectric Breakdown, spec minimum		>50	—	kV	2.5.6
Arc Resistance, spec minimum		120	60	Seconds	2.5.1
Electric Strength, spec minimum (Laminate & prepreg as laminated)		48 1200	30 750	kV/mm (V/mil)	2.5.6.2
Comparative Tracking Index (CTI)		3 (175 - 249)	-	Class (volts)	UL-746A ASTM D3638
Peel Strength, Spec Minimum	A. Low profile copper foil and very low profile – all copper weights >17 microns	6.5(1.14)	4.0(0.70)	lb/inch(N/mm)	2.4.8
	B. Standard profile copper				2.4.8.2
	1. After thermal stress	7(1.225)	4.5(0.8)		2.4.8.3
	2. At 125°C (257°F)	6.5(1.14)	4.0(0.70)		
	3. After process solutions	5.1(0.9)	3.0(0.55)		
Flexular Strength, minimum	A. Lengthwise direction	58,000	—	lb/inch²	
	B. Crosswise direction	79,000	—		
Moisture Absorption, spec maximum		0.15	—	%	2.6.2.1
Flammability (Laminate & prepreg as laminated), spec min		V0		Rating	UL-94
HWI		2			
Max Operating Temperature		130 (150)	UL Cert (tested)	Deg C	
DSR		yes			
TensileStrength, minimum	A. Lengthwise direction	50,000	—	lb/inch²	
	B. Crosswise direction	37,000	—		
Poisson's Ratio	A. Lengthwise direction	0.16	—		
	B. Crosswise direction	0.13	—		
Youngs Modulus	A. Lengthwise direction	3.4	—		
	B. Crosswise direction	3.0	—		
Taylors Modulus	A. Lengthwise direction	3.7	—		
	B. Crosswise direction	3.0	—		

The data, while believed to be accurate and based on analytical methods considered to be reliable, is for information purposes only. Any sales of these products will be governed by the terms and conditions of the agreement under which they are sold.

ORDERING INFORMATION:
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